Ruitong Tian

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Education

- **University of Florida** — *2025 – Present*, Gainesville, FL
  Ph.D. student | Sustainable Construction. Advisor: Eric Jing Du.
  Research focus: Dexterous manipulation, robot learning, and tactile sensing.
- **National University of Singapore** — *2023 – 2025*, Singapore
  M.S. | Infrastructure Digitalization & Management. Exchange student.
- **South China University of Technology (SCUT)** — *2020 – 2024*, Guangzhou, China
  B.E. | Green Intelligent Construction.

Projects & Publications

ExoBridge: Learning a Bare Hand to Hand-Worn Exoskeleton Mapping through Human Limb Coupling — **Under submission**, *University of Florida*
Ruitong Tian, Xianyao Li, Noah B. Wilson, Fang Xu, Eric Jing Du.

- Developed bare-hand RGB-to-exoskeleton mapping with MediaPipe hand/scene crops, frozen DINOv2 features, and a causal temporal model to predict motion, contact, and tactile intensity.
- Evaluated 1,215 demonstrations from 31 sessions across four tasks: contact AUROC 0.916, intensity correlation 0.790, and 44.7% lower motion error versus a zero-change reference for primary channels.

From Wearable Interfaces to Dexterous Policies: Contact Shifts and Tactile Representations — **Under submission**, *University of Florida*
Ruitong Tian, Fang Xu, Noah B. Wilson, Xianyao Li, Eric Jing Du.

- Studied wearable-interface geometry and tactile inputs for Diffusion Policy manipulation with a DINO encoder, temporal U-Net, 12-DoF XHand, and Franka Emika arm.
- Tested 12 policy–task conditions with 20 rollouts each across four tasks; spatial contact and force inputs increased pooled success from 35% to 52% versus binary contact.

Semantically Enhanced Eye-Tracking Controls in Human-Robot Interaction: A Comparison Study — **Ergonomics**, **Accepted Oct. 9, 2026**, *University of Florida*
Ruitong Tian, Hengxu You, Fang Xu, Jiahao Wu, Bowen Sun, Thanakon Uthai, Eric Jing Du.

- Co-designed a 30-participant gaze/voice/gesture study using HoloLens 2, Franka Emika, FastSAM, and RGB-D localization; curated and analyzed task performance, gaze behavior, and NASA-TLX workload.
- Measured mean step completion times of 17.10 s with gaze versus 32.78 s with hand gestures; gaze analysis used a quality-screened subset of 19 participants and 96 trials.

Nudge Before You Push: Physics-Aware Navigation via Tactile Probing — **Under submission**, *University of Florida*
Xianyao Li, Fang Xu, **Ruitong Tian**, Bowen Sun, Xiao Hu, Yang Ye, Jing Du.

- Coauthored TANav tactile-probing navigation; simulation missed-push rate was 5.5% versus 28.7% with mean peak-force thresholds at comparable low-risk operating points.

Controlling crack propagation in layered beams with architected lattice-reinforced composite interlayer designs — **Construction and Building Materials (2024)**, *SCUT*
Binglin Xie, **Ruitong Tian**, Haiyang Zhao, Tongtong Ye, Yuan Zhang, Nan Hu.

- Designed Rhino/Grasshopper lattices and FDM prototypes; validated Abaqus crack models using fracture/fatigue tests and digital image correlation.

Additional Research Experience

Machine Learning for Thermal-Image Damage Prediction — *National University of Singapore*

- Developed CNN/knowledge-transfer models in PyTorch (prediction correlation 0.98); automated 2,000+ Abaqus simulation samples and processed 150+ thermal images with Python/MATLAB.

Technical Skills

- **Programming & Robotics:** Python, PyTorch, C++, MATLAB; Diffusion Policy, DINO/DINOv2, MediaPipe, FastSAM; DexUMI, XHand, Franka Emika, HoloLens 2, tactile sensing.
- **Design & Simulation:** Rhino, Grasshopper, Abaqus, FDM 3D printing, digital image correlation, GOM Correlate, CloudCompare, Ultimaker Cura.